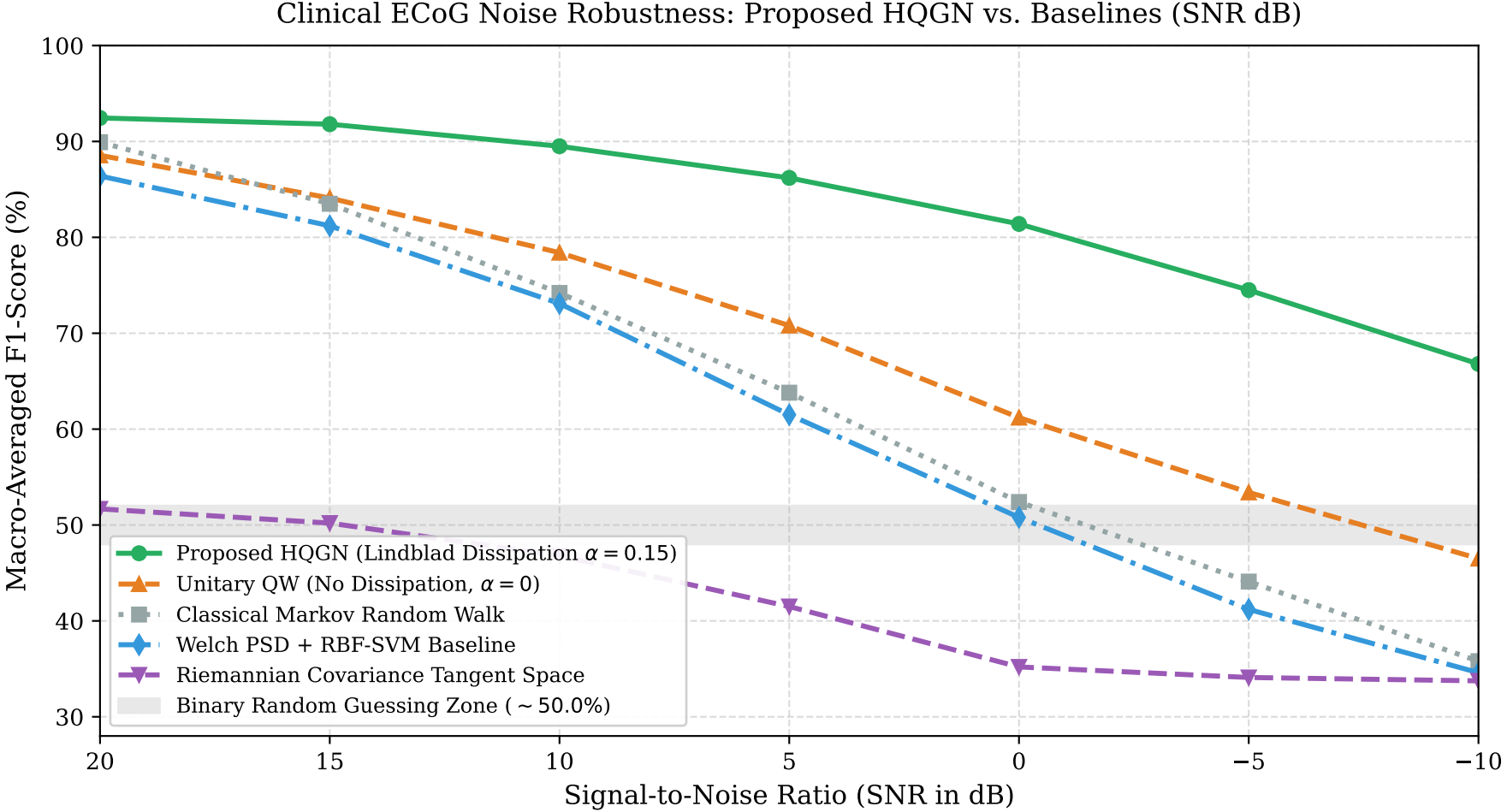
Clinical ECoG Noise Robustness: Proposed HQGN vs. Baselines (SNR dB)
Macro-Averaged F1-Score (%)
Signal-to-Noise Ratio (SNR in dB)
100
90
80
70
60
50
40
30
20
15
10
5
0
−5
−10
Proposed HQGN (Lindblad Dissipation $\alpha = 0.15$)
Unitary QW (No Dissipation, $\alpha = 0$)
Classical Markov Random Walk
Welch PSD + RBF-SVM Baseline
Riemannian Covariance Tangent Space
Binary Random Guessing Zone (~ 50.0%)